

ADVANCED

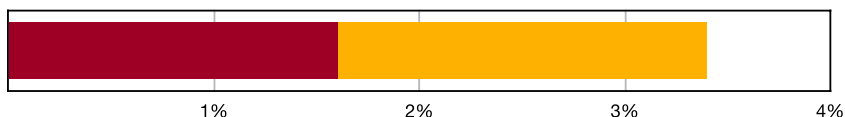
Stage 6

Calculus (Adv), C1 Introduction to Differentiation (Adv) Tangents (Y11)

Teacher: SHS Maths

Exam Equivalent Time: 51 minutes (based on allocation of 1.5 minutes per mark)

C1 Introduction to Differentiation C2 Differential Calculus



***SmarterMaths analytics based on the average contribution to new syllabus Advanced Maths exams since 2020.**

HISTORICAL CONTRIBUTION

- C1 Introduction to Differentiation* is Y11 content that has contributed an average of 1.6% per exam since the new syllabus was introduced in 2020.
- This topic has been split into three sub-topics for analysis purposes: *1-Standard Differentiation (0.8%)*, *2-Tangents (0.8%)*, and *3-Rates of Change*.
- This analysis looks at the sub-topic *Tangents*.

HSC ANALYSIS - What to expect and common pitfalls

- Tangents* in this sub-topic only include examples that require standard differentiation and no integration. Importantly, it also excludes log, exponential and trig underlyings which are covered in later year 12 sub-topics.
- Tangent* questions that require standard differentiation have been asked 3 times in the last decade, most recently in 2023 and 2019 in 3-mark questions, the latter of which caused problems (see 2019 Adv 14d).
- While regular question types offer a great opportunity for high scoring, take note of some more applied examples that created problems for many students (2009 HSC 6c, 2010 HSC 7b).

Questions

1. Calculus, 2ADV C1 EQ-Bank 3 MC

At which point on the curve $y = 2x^2 - 11x + 3$ can a tangent be drawn such that it is inclined at 45° when it crosses the positive x -axis?

- $(-3, 54)$
- $(-2, 33)$
- $(2, -11)$
- $(3, -12)$

2. Calculus, 2ADV C1 2009 HSC 1d

Find the gradient of the tangent to the curve $y = x^4 - 3x$ at the point $(1, -2)$. (2 marks)

3. Calculus, 2ADV C1 2012 HSC 11c

Find the equation of the tangent to the curve $y = x^2$ at the point where $x = 3$. (2 marks)

4. Calculus, 2ADV C1 2023 HSC 14

Find the equation of the tangent to the curve $y = (2x + 1)^3$ at the point $(0, 1)$. (3 marks)

5. Calculus, 2ADV C1 EQ-Bank 3

- Use differentiation by first principles to find y' , given $y = 2x^2 + 5x$. (2 marks)
- Find the equation of the tangent to the curve when $x = 1$. (1 mark)

6. Calculus, 2ADV C1 2011 HSC 2c

Find the equation of the tangent to the curve $y = (2x + 1)^4$ at the point where $x = -1$. (3 marks)

7. Calculus, 2ADV C1 2017 HSC 12a

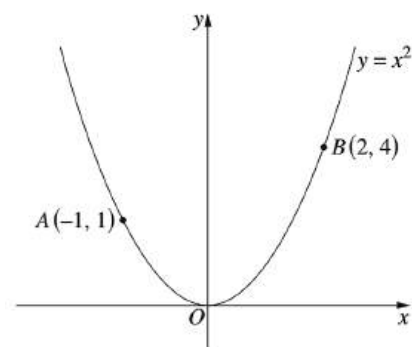
Find the equation of the tangent to the curve $y = x^2 + 4x - 7$ at the point $(1, -2)$. (2 marks)

8. Calculus, 2ADV C1 SM-Bank 2

- Find the equations of the tangents to the curve $y = x^2 - 3x$ at the points where the curve cuts the x -axis. (2 marks)
- Where do the tangents intersect? (2 marks)

9. Calculus, 2ADV C1 2010 HSC 7b

The parabola shown in the diagram is the graph $y = x^2$. The points $A(-1, 1)$ and $B(2, 4)$ are on the parabola.



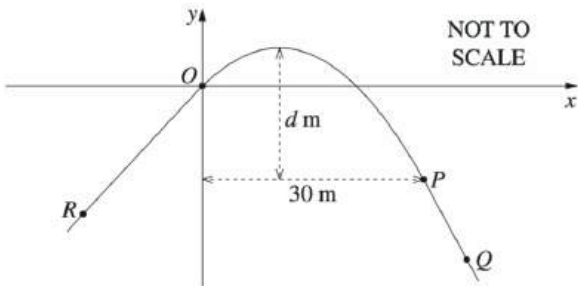
- i. Find the equation of the tangent to the parabola at A . (2 marks)
- ii. Let M be the midpoint of AB .
There is a point C on the parabola such that the tangent at C is parallel to AB .
Show that the line MC is vertical. (2 marks)
- iii. The tangent at A meets the line MC at T .
Show that the line BT is a tangent to the parabola. (2 marks)

10. Calculus, 2ADV C1 2019 HSC 14d

The equation of the tangent to the curve $y = x^3 + ax^2 + bx + 4$ at the point where $x = 2$ is $y = x - 4$.
Find the values of a and b . (3 marks)

11. Calculus, 2ADV C1 2009 HSC 6c

The diagram illustrates the design for part of a roller-coaster track. The section RO is a straight line with slope 1.2, and the section PQ is a straight line with slope -1.8 . The section OP is a parabola $y = ax^2 + bx$. The horizontal distance from the y -axis to P is 30 m.



- In order that the ride is smooth, the straight line sections must be tangent to the parabola at O and at P .
- i. Find the values of a and b so that the ride is smooth. (3 marks)
 - ii. Find the distance d , from the vertex of the parabola to the horizontal line through P , as shown on the diagram. (2 marks)

Worked Solutions

1. Calculus, 2ADV C1 EQ-Bank 3 MC

$$y = 2x^2 - 11x + 3$$

$$y' = 4x - 11$$

If a tangent crosses (positive) x-axis at 45° ,

$$m_{\text{tang}} = 1 \quad (\tan 45^\circ = 1)$$

Find x when $y' = 1$:

$$4x - 11 = 1$$

$$4x = 12$$

$$x = 3$$

Tangent at point $(3, -12)$

$$\Rightarrow D$$

2. Calculus, 2ADV C1 2009 HSC 1d

$$y = x^4 - 3x$$

$$\frac{dy}{dx} = 4x^3 - 3$$

$$\text{At } x = 1$$

$$\frac{dy}{dx} = 4 - 3 = 1$$

$\therefore$ Gradient of tangent at $(1, -2) = 1$.

Worked Solutions

3. Calculus, 2ADV C1 2012 HSC 11c

$$y = x^2$$

$$\frac{dy}{dx} = 2x$$

Need to find equation with $m = 6$, through $(3, 9)$

$$\text{When } x = 3, y = 9, \frac{dy}{dx} = 6$$

$$\text{Using } y - y_1 = m(x - x_1)$$

$$y - 9 = 6(x - 3)$$

$$y - 9 = 6x - 18$$

$$6x - y - 9 = 0$$

$$\therefore \text{ Equation of the tangent is } 6x - y - 9 = 0$$

4. Calculus, 2ADV C1 2023 HSC 14

$$y = (2x + 1)^3$$

$$\frac{dy}{dx} = 3 \times 2(2x + 1)^2$$

$$= 6(2x + 1)^2$$

$$\text{At } x = 0 \Rightarrow \frac{dy}{dx} = 6 \times 1^2 = 6$$

Find equation of line $m = 6$, through $(0, 1)$

$$y - y_1 = m(x - x_1)$$

$$y - 1 = 6(x - 0)$$

$$y = 6x + 1$$

5. Calculus, 2ADV C1 EQ-Bank 3

i. $f(x) = 2x^2 + 5x$

$$\begin{aligned} f'(x) &= \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} \\ &= \lim_{h \rightarrow 0} \frac{(2(x+h)^2 + 5(x+h)) - (2x^2 + 5x)}{h} \\ &= \lim_{h \rightarrow 0} \frac{2x^2 + 4xh + 2h^2 + 5x + 5h - 2x^2 - 5x}{h} \\ &= \lim_{h \rightarrow 0} \frac{4xh + 2h^2 + 5h}{h} \\ &= \lim_{h \rightarrow 0} \frac{h(4x + 5 + 2h)}{h} \end{aligned}$$

$\therefore y' = 4x + 5$

ii. When $x = 1, y = 7$

$y' = 4 + 5 = 9$

$\therefore y - 7 = 9(x - 1)$
 $y = 9x - 2$

6. Calculus, 2ADV C1 2011 HSC 2c

$y = (2x + 1)^4$
Using the Function of a Function Rule (or Chain Rule)

$$\begin{aligned} \frac{dy}{dx} &= 4 \times (2x + 1)^3 \times \frac{d}{dx}(2x + 1) \\ &= 8(2x + 1)^3 \end{aligned}$$

At $x = -1, y = 1$

$$\begin{aligned} \frac{dy}{dx} &= 8(2(-1) + 1)^3 \\ &= 8(-1)^3 \\ &= -8 \end{aligned}$$

Tangent has $m = -8$ through $(-1, 1)$

Using $y - y_1 = m(x - x_1)$
 $y - 1 = -8(x + 1)$
 $y - 1 = -8x - 8$
 $8x + y + 7 = 0$

$\therefore$ Equation of tangent is $8x + y + 7 = 0$

MARKER'S COMMENT: The best setting out clearly showed the derivative function, the gradient, the point and then finally, calculations for the equation of the tangent, as per the Worked Solution.

7. Calculus, 2ADV C1 2017 HSC 12a

$y = x^2 + 4x - 7$
 $\frac{dy}{dx} = 2x + 4$

When $x = 1, \frac{dy}{dx} = 6$

Equation of tangent through $(1, -2)$

$y + 2 = 6(x - 1)$
 $y = 6x - 8$

8. Calculus, 2ADV C1 SM-Bank 2

i. $y = x^2-3x$
 $= x(x-3)$

Cuts x -axis at $x = 0$ or $x = 3$

$$\frac{dy}{dx} = 2x-3$$

At $x = 0 \Rightarrow \frac{dy}{dx} = -3$

T_1 has $m = -3$, through $(0, 0)$

$$y-0 = -3(x-0)$$
$$y = -3x$$

At $x = 3 \Rightarrow \frac{dy}{dx} = 3$

T_2 has $m = 3$, through $(3, 0)$

$$y-0 = 3(x-3)$$
$$y = 3x-9$$

ii. Intersection occurs when:

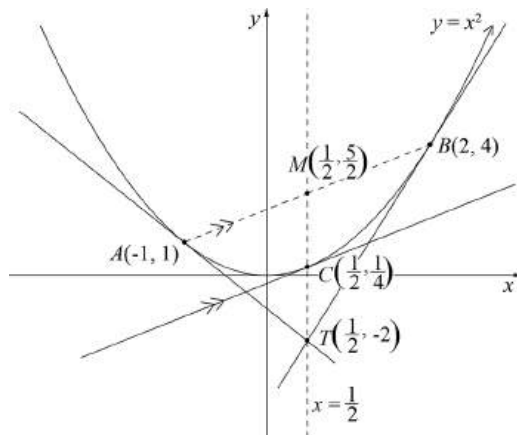
$$3x-9 = -3x$$
$$6x = 9$$
$$x = \frac{3}{2}$$

$$y = -3 \times \frac{3}{2} = -\frac{9}{2}$$

$\therefore$ Intersection at $\left(\frac{3}{2}, -\frac{9}{2}\right)$

9. Calculus, 2ADV C1 2010 HSC 7b

i.



$$y = x^2$$

$$\frac{dy}{dx} = 2x$$

$$\text{At } A(-1,1) \Rightarrow \frac{dy}{dx} = -2$$

Tangent has $m = -2$, through $(-1,1)$:

$$y - y_1 = m(x - x_1)$$

$$y - 1 = -2(x + 1)$$

$$y - 1 = -2x - 2$$

$$2x + y + 1 = 0$$

$$\therefore \text{ Tangent at } A \text{ is } 2x + y + 1 = 0$$

ii. $A(-1,1)$ $B(2,4)$

$$M = \left(\frac{-1 + 2}{2}, \frac{1 + 4}{2} \right)$$

$$= \left(\frac{1}{2}, \frac{5}{2} \right)$$

$$m_{AB} = \frac{y_2 - y_1}{x_2 - x_1}$$

$$= \frac{4 - 1}{2 - (-1)} = 1$$

$$\text{When } \frac{dy}{dx} = 1$$

$$2x = 1$$

$$x = \frac{1}{2}$$

$$\therefore C \left(\frac{1}{2}, \frac{1}{4} \right)$$

$$\Rightarrow M \text{ and } C \text{ both have } x\text{-value} = \frac{1}{2}$$

$$\therefore MC \text{ is vertical } \dots \text{ as required}$$

iii. T is point on tangent when $x = \frac{1}{2}$

$$\text{Tangent } 2x + y + 1 = 0$$

$$\text{At } x = \frac{1}{2}$$

$$2 \times \left(\frac{1}{2} \right) + y + 1 = 0$$

$$\Rightarrow y = -2$$

$$\therefore T \left(\frac{1}{2}, -2 \right)$$

$$\text{Given } B(2,4)$$

$$m_{BT} = \frac{4 - (-2)}{2 - \frac{1}{2}}$$

$$= 4$$

At $B(2,4)$, find gradient of tangent:

$$\frac{dy}{dx} = 2x = 2 \times 2 = 4$$

$$\therefore m_{\text{tangent}} = 4 = m_{BT}$$

$$\therefore BT \text{ is a tangent}$$

♦♦ Mean mark 29%.

♦ Mean mark 37%.
IMPORTANT: The critical understanding required for this question is that the gradient of AB needs to be equated to the gradient function (i.e. $\frac{dy}{dx}$).

10. Calculus, 2ADV C1 2019 HSC 14d

$$y = x^3 + ax^2 + bx + 4$$
$$\frac{dy}{dx} = 3x^2 + 2ax + b$$

When $x = 2$, $\frac{dy}{dx} = 1$

$$12 + 4a + b = 1$$
$$4a + b = -11 \dots (1)$$

The point $(2, -2)$ lies on y :

$$8 + 4a + 2b + 4 = -2$$
$$4a + 2b = -14 \dots (2)$$

Subtract $(2)-(1)$

$$b = -3$$

Substitute into (1)

$$4a - 3 = -11$$
$$4a = -8$$
$$a = -2$$

◆ Mean mark 46%.

11. Calculus, 2ADV C1 2009 HSC 6c

i. $y = ax^2 + bx$

$$\frac{dy}{dx} = 2ax + b$$

At $x = 0$,

$$\frac{dy}{dx} = b$$

We need m at $O = 1.2$

$$\therefore b = 1.2$$

At P , $x = 30$

$$\frac{dy}{dx} = 2 \times a \times 30 + 1.2$$
$$= 60a + 1.2$$

We need m at $P = -1.8$

$$60a + 1.2 = -1.8$$
$$60a = -3$$
$$a = -\frac{3}{60}$$
$$= -0.05$$

$$\therefore \text{ For a smooth ride, } a = -0.05 \text{ and } b = 1.2$$

ii. $y = -0.05x^2 + 1.2x$

$$\frac{dy}{dx} = -0.1x + 1.2$$

Find x when $\frac{dy}{dx} = 0$

$$-0.1x + 1.2 = 0$$
$$x = \frac{1.2}{0.1}$$
$$= 12$$

$$\therefore \text{ MAX when } x = 12$$

When $x = 12$

$$y = -0.05 \times 12^2 + 1.2 \times 12$$
$$= -7.2 + 14.4$$
$$= 7.2$$

When $x = 30$

◆◆◆ *Exact data unavailable.*
MARKE R'S COMMENT: Successful students equated the curve gradient to the straight section as a requirement for a smooth ride.

$$\begin{aligned}y &= -0.05 \times 30^2 + 1.2 \times 30 \\&= -45 + 36 \\&= -9\end{aligned}$$

$$\begin{aligned}\therefore d &= 7.2 + |-9| \\&= 16.2 \text{ m}\end{aligned}$$